

## **CSA1717-ARTIFICIAL INTELLIGENCE PRACTICAL LAB EXPERIMENTS**

### **PLANETS DATABASE**

**Aim:** create a Prolog database containing information about planets and retrieve their properties.

**Objective:**

To represent planetary information using facts and retrieve planets based on their properties.

**Algorithm: Start.**

**Define planet facts.**

**Store properties such as type and order. Query the database.**

**Display matching planets. Stop.**

**Flowchart:**

START

| v

Create Planet Database

| v

Enter Property/Planet

| v

Search Database

| v

Display Matching Planet

| v

STOP

**Prolog Program:**

**% Planets Database**

**planet(mercury, terrestrial, 1).**

**planet(venus, terrestrial, 2).**

**planet(earth, terrestrial, 3).**

**planet(mars, terrestrial, 4).**  
**planet(jupiter, gas\_giant, 5).**  
**planet(saturn, gas\_giant, 6).**  
**planet(uranus, ice\_giant, 7).**  
**planet(neptune, ice\_giant, 8).**

**terrestrial(P) :-**

**planet(P, terrestrial, \_).**

**gas\_giant(P) :-**

**planet(P, gas\_giant, \_).**

**ice\_giant(P) :-**

**planet(P, ice\_giant, \_).**

**Output:**

**Result:**

Thus, the Prolog program successfully creates and queries a planets database.